



Evaluating the efficacy of an innovative herbal formulation (HF6) on different human cancer cell lines

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Abstract

Cancer is reported to be the leading cause of death and illness worldwide. This research aims to evaluate the phytochemicals, antioxidant, cytotoxic, and apoptotic activities of the polyherbal formulation HF6. HF6 was prepared by blending equal quantities of plants powder, namely, *Curcuma longa*, *Salvia officinalis*, *Cinnamomum zeylanicum*, *Capsicum annuum*, *Zingiber officinale*, and *Syzygium aromaticum*, and later extracted using hexane (HF6H), chloroform (HF6C), ethyl acetate (HF6E), and methanol (HF6M) in Soxhlet apparatus. Among the four different extracts, only the hexane extract (HF6H) was significantly effective. The HF6H extract showed antioxidant and anticancer potentials against different cancer cell lines, and moderate cytotoxicity against non-cancer cells, rendering it a promising remedy. In addition, it exerted tremendous cytotoxic effects on MCF-7, Huh-7, HCT116, MDA-MB-231, LoVo, and HepG2 cells with IC₅₀ values of 2.02, 4.5, 6.9, 11.4, 23.5, and 34.7 µg/mL, respectively. The morphological hallmarks of apoptosis such as the rounding of cells, loss of contact with neighboring cells, formation of cell membrane blebbing, and microspike protrusion were detected using several different techniques. DAPI staining revealed apoptotic nuclear morphology such as condensation and DNA fragmentation. The morphological changes of MCF7 cells were also analyzed by AO/EB fluorescence staining. MCF7-stained green cells were viable cells, whereas the treated cells showed fragmented green nuclei representing early apoptosis. The phytochemical screening of HF6H showed positive results regarding the presence of alkaloids, polyphenols, flavonoids, and sterols. The GC–MS (gas chromatography–mass spectrometry) analysis of the HF6H extract indicated the presence of 12 compounds, mainly trans-caryophyllene (21.55%), cis-isoeugenol (18.42%), acetyleugenol (17.53%), alpha farnesene (10.0%), and zingiberene (8.55%). However, further investigation could be carried out to examine the toxicity of the extract on animal models.

Keywords Polyherbal · Anticancer · Apoptosis · GC–MS · Antioxidant

Introduction

Cancer is considered as the leading cause of death and illness worldwide regardless of preventive measures, advanced diagnosis tools, and treatment methods (El-Sayed et al. 2021; He et al. 2016). The uncontrolled growth of normal cells due to internal factors (genetic mutations, body immune system, and hormonal disorders) and external factors (radiations, smoking, tobacco, pollutants, chemicals, and infectious agents) (Wu et al.

2018) can produce genetic instabilities that transform normal cells into malignant cells. These genetic instabilities include mutations in oncogenes, tumor suppressor genes, DNA repair genes, and genes involved in cell growth and metabolism (Singh et al. 2016).

There are many therapeutic strategies to treat cancer (Blumen et al. 2016). However, these methods are often accompanied by different side effects (Love et al. 1989). Irinotecan, doxorubicin, carboplatin, cisplatin, vincristine, vinblastine, docetaxel, and paclitaxel are effective anti-cancer agents against a wide variety of cancers (Lichota & Gwozdinski 2018). Nevertheless, there are some limitations associated with these chemotherapeutic agents like side effects, high-cost, toxic reactions, and the development of cell resistance (Nurgali et al. 2018). Studies have shown the ability of cancer cells in resisting several chemotherapeutic agents including the recently developed

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anticancer drugs: abemaciclib, doxorubicin, 5-fluorouracil, paclitaxel, and vinblastine (AbuHammad & Zihlif 2013, Guarducci et al. 2017, Němcová-Fürstová et al. 2016, Rosas-Ramírez et al. 2017).

Plant extracts serve as an available source for the treatment of chronic and non-chronic diseases. In addition, they are tested for their beneficial health effects by many researchers (AlBasher et al. 2020; Kandeil et al. 2020; Mohammed et al. 2020; Mohammed & Safwat 2020; Singh et al. 2021).

There is a demand for novel and safe anticancer drugs for the treatment of cancer. Interestingly, there has been an increase in the use of polyherbal formulations as an adjunct to cancer drugs (Abutaha et al. 2021; Chen et al. 2014; Hussain et al. 2016). These remedies have an anticancer potential and could enhance the immune system (Jeba et al. 2011; Murunikkara & Rasool 2014). In addition, they have been recognized to improve the survival rates of patients, reduce the side effects, and chemotherapy-associated toxicity (Kummar et al. 2011). Therefore, the present research was carried out to develop an innovative polyherbal formulation capable of inducing cytotoxicity in different cancer cells.

Materials and methods

Plant collections

The *Curcuma longa* L. (voucher no. BPR/101) rhizome, *Salvia officinalis* L. (voucher no. BPR/102) leaves, *Cinnamomum zeylanicum* bark Blume (voucher no. BPR/103), *Capsicum annuum* L. fruit (voucher no. BPR/104), *Zingiber officinale* rhizome L. (voucher no. BPR/105), and *Syzygium aromaticum* L. seeds (voucher no. BPR/106) were purchased from a herbal drug store. The plants were authenticated, and voucher samples were placed at the Botany Department and the Bio-Product Research Chair of the College of Science at King Saud University.

Plant extraction

The herbal formula was composed of equal quantities (2 g each) of the selected medicinal plants grounded using a commercial blender and filled in a thimble. The thimble was placed in the Soxhlet extractor and extracted using HPLC grade (Sigma, USA) hexane (HF6H), chloroform (HF6C), ethyl acetate (HF6E), and methanol (HF6M), respectively. The collected extract from each solvent was evaporated under a vacuum at 45 °C (Heidolph,

Germany). The extract was dissolved in dimethyl sulfoxide (DMSO).

Phytochemical tests

The phytochemical tests were carried out using HF6H extract. The screening was performed as reported previously (Iqbal et al. 2015). Detection of alkaloids, saponins, steroids/triterpenoids, flavonoids, and tannins/polyphenols was assessed using Dragendorff and Mayer's tests, foam appearance, Liebermann-Buchard's test, ferric chloride test, and Shinoda's test, respectively.

DPPH free radical scavenging assay

The antiradical activity was assessed using a UV–visible spectrophotometer at 520 nm (Abutaha et al. 2021). The reaction mixtures consist of 5 µL of HF6H extract and 195 µL of 0.004% DPPH·(Sigma, USA) dissolved in methanol. Different concentrations (5 to 0.07 mg/mL) of the sample were incubated for 30 min at 25 °C. DMSO (Panreac, Spain) was used as a blank; each experiment was measured in triplicate. The DPPH activity was expressed as a percentage and calculated using the equation below:

$$\text{Scavenging\%} = \frac{A_{\text{control}} - A_{\text{extract}}}{A_{\text{control}}} \times 100$$

Gas chromatography–mass spectrometry (GC–MS)

The phytochemical analysis of HF6H extract was investigated using GC–MS (Agilent Technologies, USA), as reported earlier (Hidayathulla et al. 2018). Identification of compounds was accomplished by comparing their mass spectra to the National Institute of Standards and Technology (NIST) library (NIST 2008).

Cell culture

Human hepatocellular carcinoma (CHANG, Huh-7, HepG2), human colon carcinoma (LoVo, HCT116), and human breast carcinoma (MCF7 and MDA-MB-231) were obtained from the German Collection of Cell Cultures (DSMZ). Also, the normal human umbilical vein endothelial cells (HUVECs) were obtained from the Japanese Collection of Research Bioresources.

Cell proliferation assays

MTT (3-(4,5-dimethylthiazol-2-yl)-2,5-diphenyltetrazolium bromide, a tetrazole) (Thermo Fisher, USA) assay

was employed to assess the cellular proliferation as previously reported (Abutaha et al. 2021). Cancer cells (5×10^5) were grown in 24 well plates, and extracts were added at increasing concentrations (10–100 $\mu\text{g/mL}$) over a 48-h period. Cells were grown at 37 °C in a 5% CO_2 humidified incubator (Binder, Germany). Also, toxicity was read at 570 nm using a spectrophotometer (ChroMate, England). Experiments were conducted in triplicate for each cell line. DMSO (1%) and cyanide 3-chlorophenylhydrazine were used as negative and positive controls, respectively.

Morphological changes

Morphological changes were evaluated in the absence or presence of extract ($2 \times \text{IC}_{50}$) concentrations. The cells were cultured and treated as mentioned earlier. At the end of the treatment period, cells were imaged ($200\times$) using a phase-contrast microscope (EVOS, USA). DMSO (1%) was used as negative control.

DAPI staining

Changes in cell nuclear morphology were assessed by DAPI (Thermo Fisher, USA) staining (Abutaha et al. 2021). The cells were treated as stated in the previous section and later washed using PBS (phosphate buffered saline), fixed with cold methanol (20 min), and stained with 1 $\mu\text{g/mL}$ DAPI at 25 °C for 15 min, sequentially. DMSO (1%) was used as negative control. Finally, all the wells were washed with PBS, and cells were imaged using a fluorescent microscope at $200\times$ magnification.

Acridine orange/ethidium bromide (AO/EB) staining

The cells were cultured and treated as described earlier. Changes in the morphology of the cells were assessed using AO/EB fluorescent staining (Abutaha et al. 2021). Briefly, the extract-treated cells were stained with 1 $\mu\text{L/mL}$ of AO/EB mixture, containing 5 $\mu\text{g/mL}$ AO and 5 $\mu\text{g/mL}$ EB (Sigma, USA). DMSO (1%) was used as negative control. After staining, cells were imaged using a fluorescent microscope at $200\times$ magnification.

Statistical analysis

The results were expressed as means $\pm$ SD of 3 replicates per experiment. Prism 9 software was used to calculate the IC_{50} values using non-linear regression analysis.

Result

The cytotoxic effects of the HF6H, HF6C, HF6E, and HF6M extracts were evaluated against HepG2, Huh-7, CHANG, LoVo, HCT116, MCF7, and MDA-MB-231 cells. Only the HF6H extract was active against all the cells examined, and the IC_{50} values were determined for all the cell lines except CHANG, which had an IC_{50} value higher than 100 $\mu\text{g/mL}$. The HF6H extract showed dose-dependent toxicity, as represented in Figs. 1 and 2. The most potent cytotoxic activities of the HF6H extract were against MCF7, Huh-7, HCT116, MDA-MB-231, LoVo, and HepG2 cells with IC_{50} values of 2.02, 4.5, 6.9, 11.4, 23.5, and 34.7 $\mu\text{g/mL}$, respectively. The most sensitive cell line tested was MCF7, followed by Huh-7. Carbonyl cyanide 3-chlorophenylhydrazine was used as a positive control, and its corresponding IC_{50} values were 29.0, 18.3, 2.2, 1.3, 10.6, and 53.2 against LoVo, MCF7, MDA-MB-231, Huh-7, HCT116, and HepG2, respectively. Its highest cytotoxic activity was against the Huh-7 cell line, followed by MDA-MB-231. Cytotoxicity evaluation against the normal HUVEC cells revealed that HF6H possessed an IC_{50} value of 55.45 $\mu\text{g/mL}$, whereas the IC_{50} value of the cyanide 3-chlorophenylhydrazine was 44.98 $\mu\text{g/mL}$ (Figs. 1 and 2).

Effects of HF6H extract on MCF-7 morphology

Direct observation of the MCF7 cells revealed a decrease in the number of cells and growth inhibition after treatment with HF6H extract. The observed morphological changes showed the formation of cell membrane blebbing, microspike protrusion, rounding of cells, and detachment from neighboring cells which confirmed the occurrence of apoptosis. The control cells (0.01% DMSO) showed higher confluency than HF6H-treated cells (Fig. 3A). MCF7 cells were stained with DAPI and AO/EB dye post-incubation with HF6H extract ($2 \times \text{IC}_{50}$) to validate that the cytotoxicity observed post-treatment with HF6H is due to apoptosis. As shown in Fig. 3, DAPI staining revealed typical apoptotic nuclear morphology such as nuclear shrinkage, DNA fragmentation, and condensation. However, these observations were not detected in control cells (0.01% DMSO) (Fig. 3C). The morphological changes of the HF6H-treated MCF7 cells were also analyzed by AO/EB fluorescent staining. Results revealed more apoptotic cells post-treatment with HF6H than the control cells. Cells treated with HF6H showed characteristic features of apoptosis such as nuclear condensation, fragmentation, cell shrinkage, and formation of apoptotic bodies. Cells treated with $2 \times \text{IC}_{50}$ of HF6H showed early apoptosis (fragmented green nucleus) (Fig. 4). MCF7-stained green cells are viable cells.

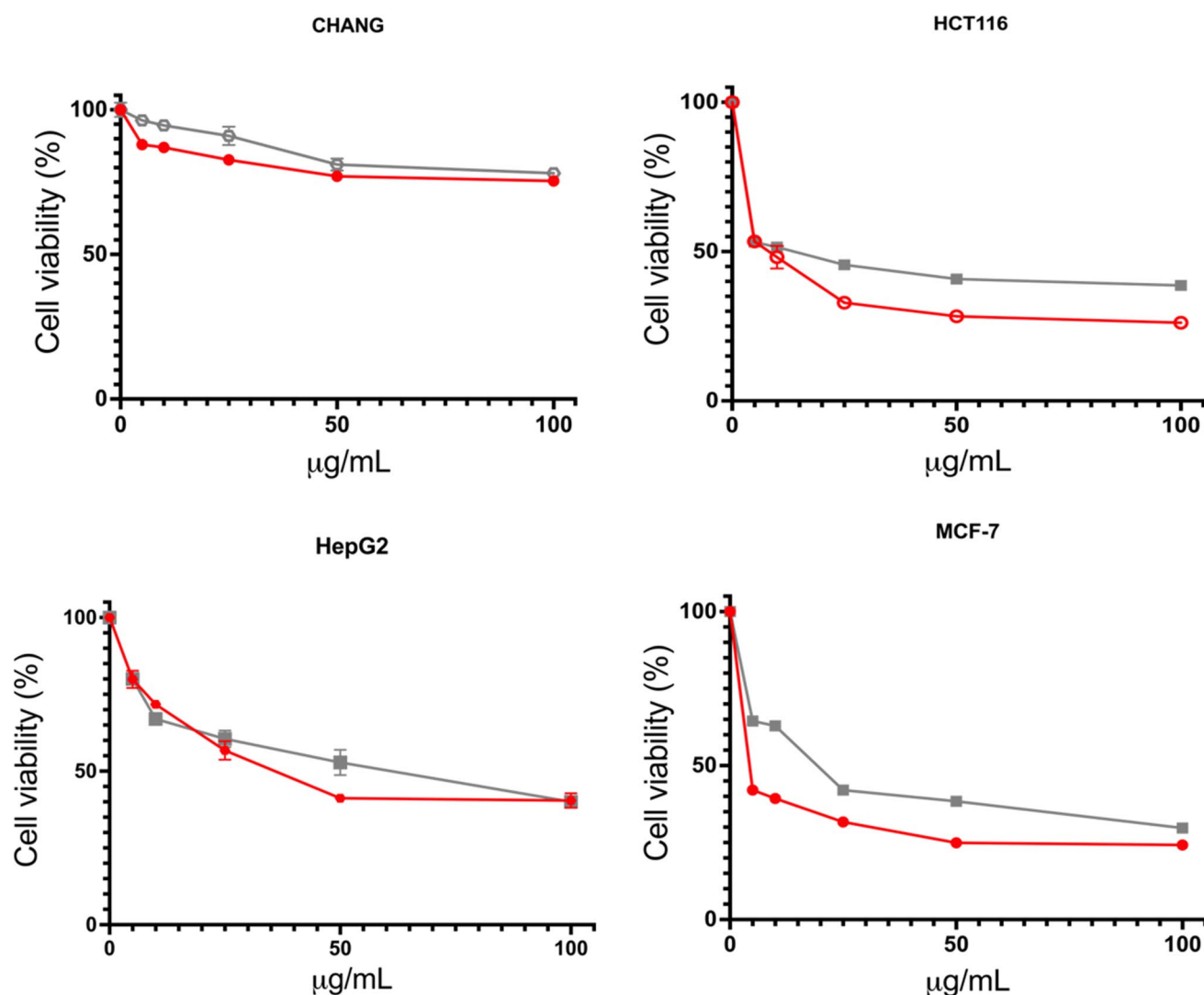


Fig. 1 Cytotoxicity of HF6H extract and cyanide 3-chlorophenylhydrazine (positive control) against human breast carcinoma (MCF7), human colon carcinoma (LoVo), human hepatocellular carcinoma (HepG2, CHANG), and the normal human umbilical vein endothelial

cells (HUVECs) using MTT assay after 24 h of treatment with different concentrations. Circle: HF6H extract, square: cyanide 3-chlorophenylhydrazine. Values are mean $\pm$ standard deviation of triplicate determinations

Antioxidant activity

The radical-scavenging abilities of the HF6H extract showed a dose-dependent decrease in the concentration of DPPH due to the scavenging activity of the extract (Fig. 5). The EC_{50} value obtained was 3.5 mg/mL.

Phytochemical screening

The HF6H developed orange or yellow precipitate complexes when tested for the presence of alkaloid compounds using Dragendorff's and Mayer's tests, respectively. The HF6H developed reddish-brown color when

tested for the presence of sterols. Also, flavonoids were stained in yellowish color by Shinoda test. In addition, the ferric chloride test showed positive results for tannins in the extract. The HF6H showed a greenish color and white precipitate, indicating the presence of tannins. The amino acid, anthraquinone glycoside, and saponin tests showed negative results, indicating the absence of these compounds.

Gas chromatography–mass spectrometry (GC–MS)

The GC–MS analysis of the HF6H extract showed the presence of 12 phytochemicals. The predominant ones were

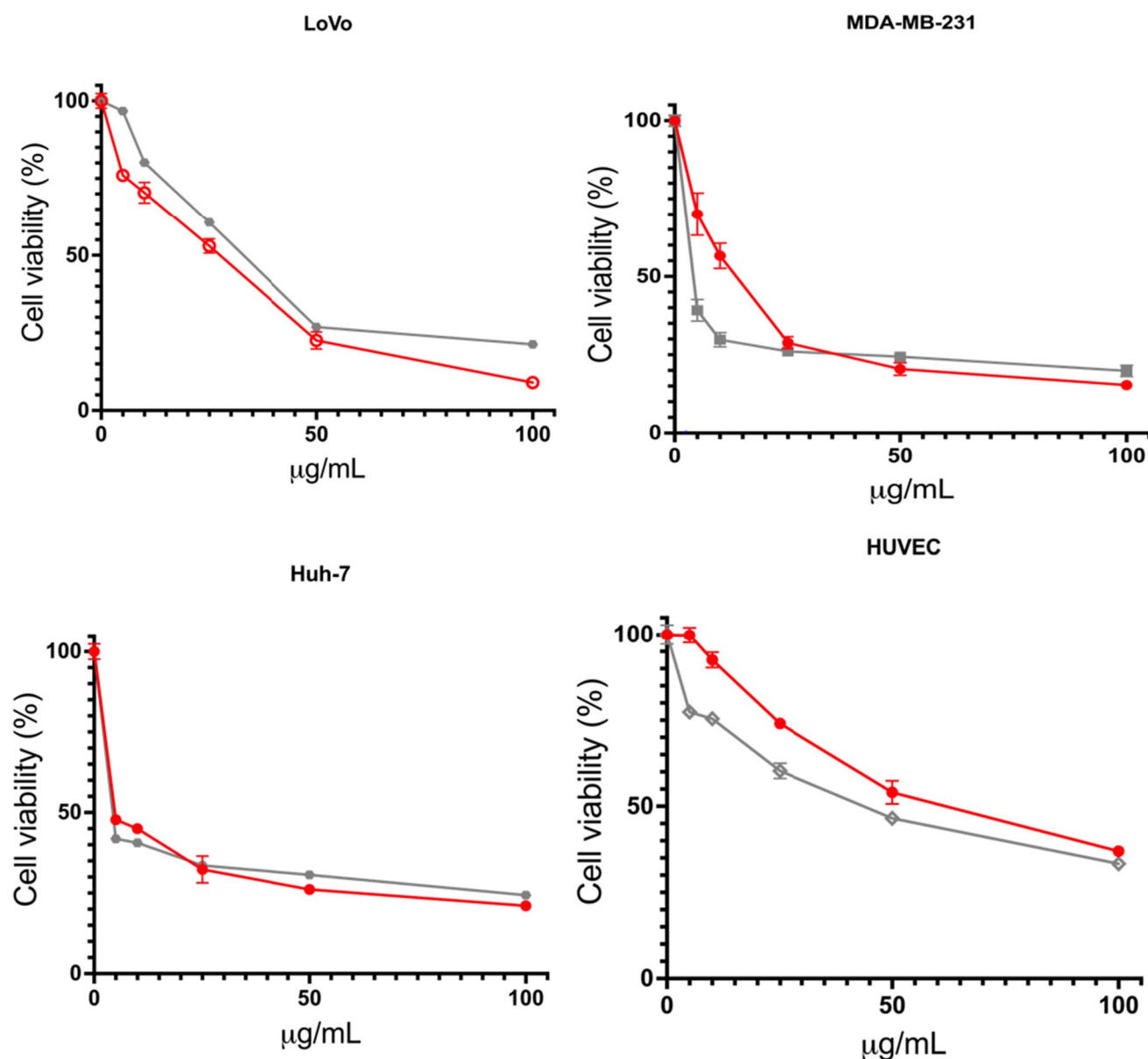


Fig. 2 Cytotoxicity of HF6H extract and cyanide 3-chlorophenylhydrazine against human breast carcinoma (MDA-MB-231), human colon carcinoma (HCT116), human hepatocellular carcinoma (Huh-7), and the normal human umbilical vein endothelial cells (HUVECs)

using MTT assay after 24 h of treatment with different concentrations. Circle: HF6H extract, square: cyanide 3-chlorophenylhydrazine. Values are mean $\pm$ standard deviation of triplicate determinations

trans-caryophyllene (21.55%), cis-isoeugenol (18.42%), acetyl-leugenol (17.53%), alpha farnesene (10.0%), and zingiberene (8.55%) (Fig. 6; Table 1).

Discussion

The cytotoxic activity and apoptotic potential were examined by evaluating the cell morphology and cytotoxicity of MCF-7 cells incubated with HF6H. Our results revealed cytotoxic potential against MCF-7 cells that caused a

reduction in the number of cells and morphological alternations associated with apoptosis. Anticancer drugs that can induce selective apoptosis, with less side effects on normal cells, are strongly recommended for greater therapeutic benefits (Buolamwini 1999; Safwat et al. 2021). The cytotoxicity evaluation against normal HUVECs revealed that HF6H exhibited less cytotoxicity on these cells. The low cytotoxic activity of HF6H against HUVEC cells, and the potent cytotoxic effects it exerted on MCF-7 cells in comparison with the carbonyl cyanide 3-chlorophenylhydrazine, suggests

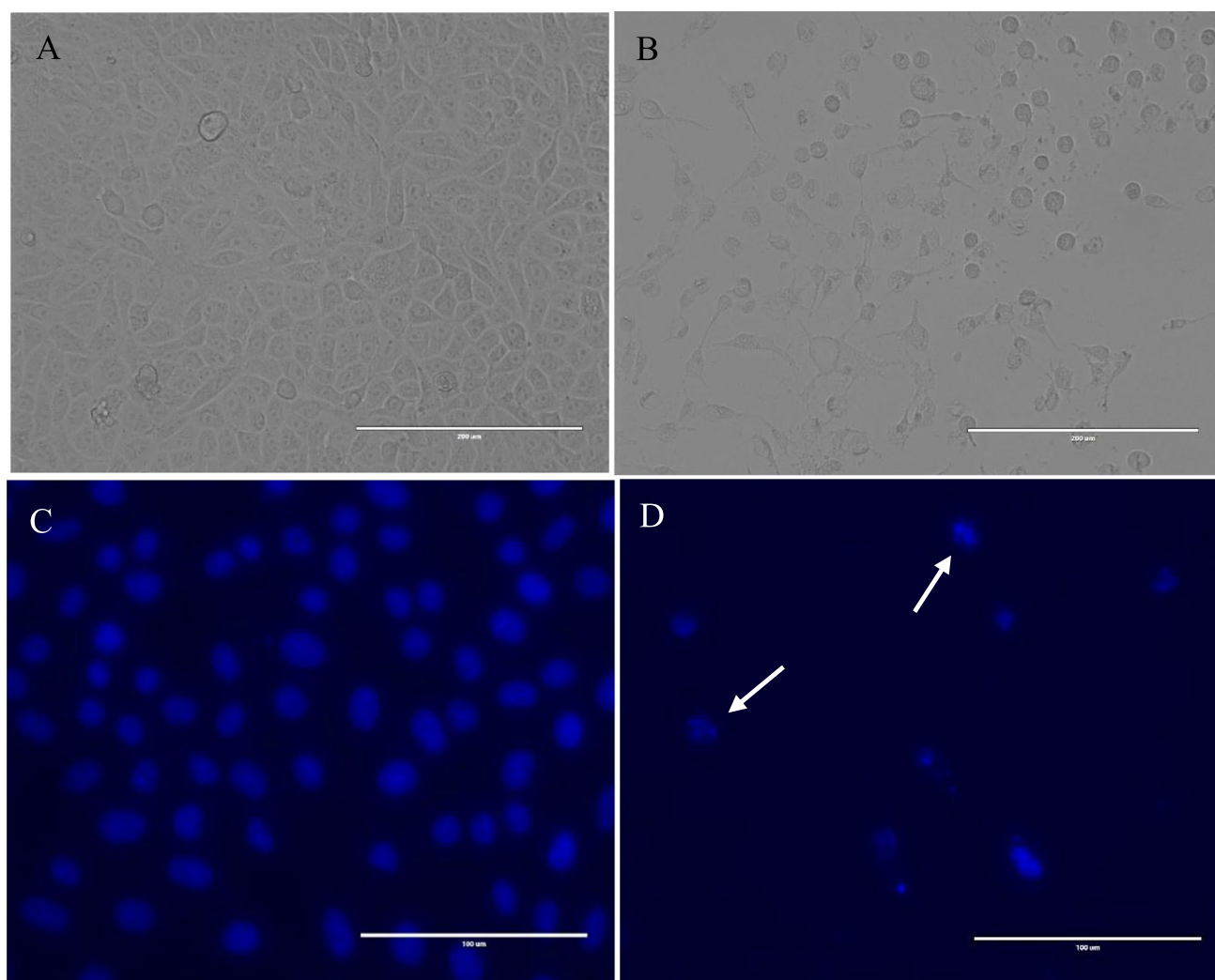


Fig. 3 Morphological assessment of MCF-7 cells treated with HF6H extract ($2 \times IC_{50}$) and observed using a phase-contrast microscope (A: control, B: treated) after 24 h of treatment. MCF7 cells with apoptotic

nuclei treated with HF6H stained by DAPI and observed using fluorescence microscope (C: control, D: treated). White arrows indicate chromatin condensation

that HF6H is more selective towards MCF-7 cells than the carbonyl cyanide 3-chlorophenylhydrazine.

Cell lines derived from the same or different organs responded differently to HF6H extract. Such outcomes are attributed to the fact that cancer cells of the same type consist of other cells that respond differently to chemotherapeutic drugs. The different responses of cell to different anticancer drugs have been reported, and it is known that different cell lines vary in their drug resistance capability (Mhaidat et al. 2014; Nannizzi et al. 2010).

MCF-7 cells treated with HF6H showed the typical characteristics of apoptosis such as cell membrane blebbing, microspike protrusion, rounding of cells, and detachment from neighboring cells. Similar results were observed when MCF-7 were treated with rusvinoxidase (Mukherjee et al. 2015), staurosporine (Mooney et al. 2002), daidzein

(Jin et al. 2002), govaniadine (Sivakumaran et al. 2018), fucoidan (Yamasaki-Miyamoto et al. 2009), and mitomycin (Pirnia et al. 2002).

Secondary metabolites found in the HF6H extract such as tannins, flavonoids, alkaloids, sterols, and terpenes are known for their antifungal, antioxidant, antiinflammatory, antiviral, and anticancer activities (Seca & Pinto 2018). Generally, secondary metabolites are reported to inhibit the occurrence and progression of cancer (Shin et al. 2018). These metabolites play vital roles in protein activation, stimulation of signaling pathways, formation of protective enzymes (caspases), activation of DNA repair mechanisms, and activation of antioxidant enzymes, thus causing anticancer effects (Barrios-González 2018). All the compounds detected using GC–MS in the HF6H extract have been reported to possess anticancer potential against different

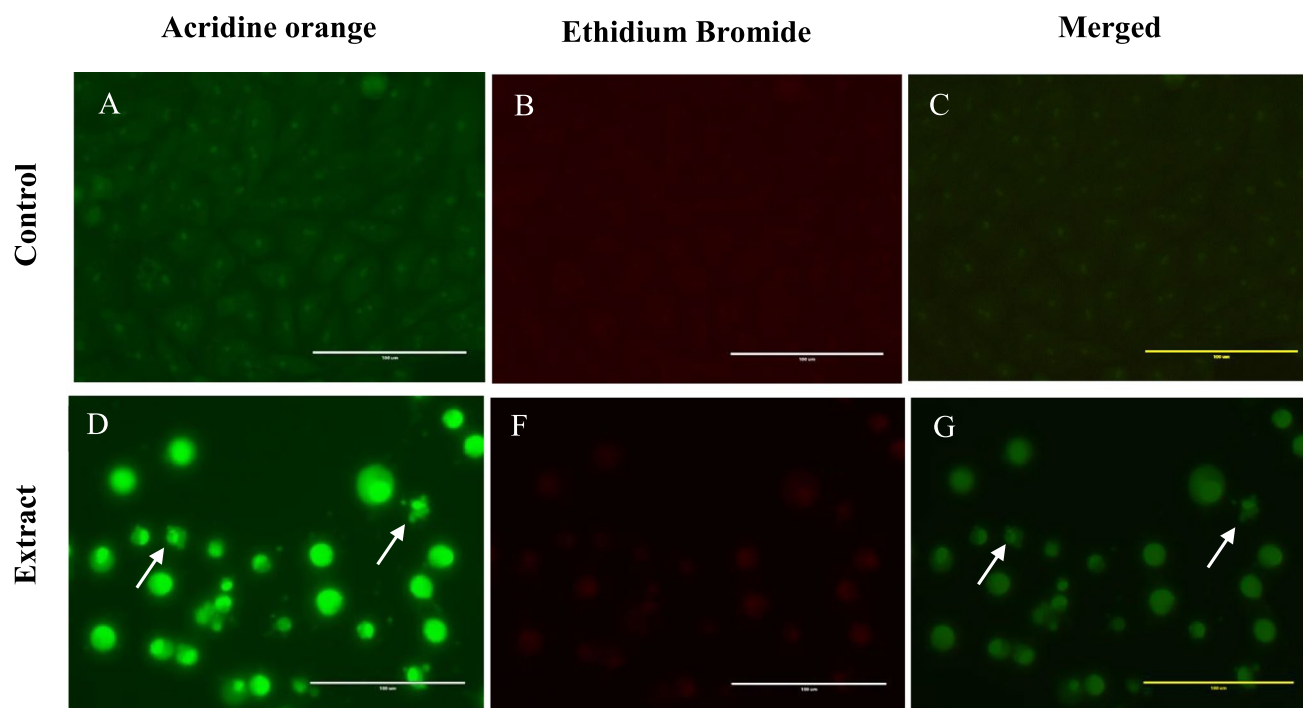


Fig. 4 Acridine orange/ethidium bromide-stained images of MCF-7 cells treated with HF6H ($2 \times IC_{50}$); arrows indicate membrane blebbing, early apoptotic cell, and fragmented nucleus

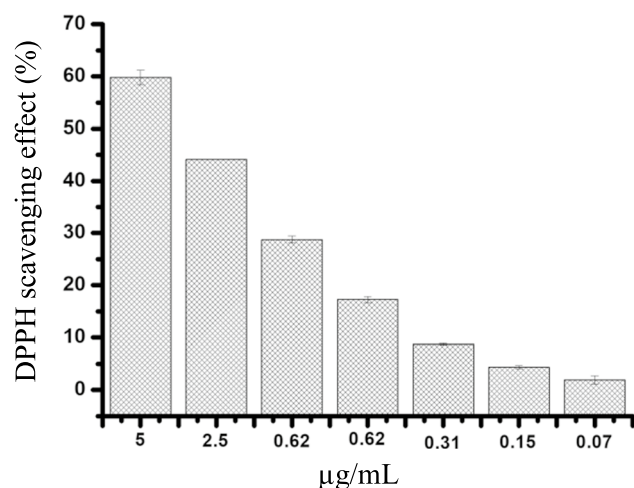
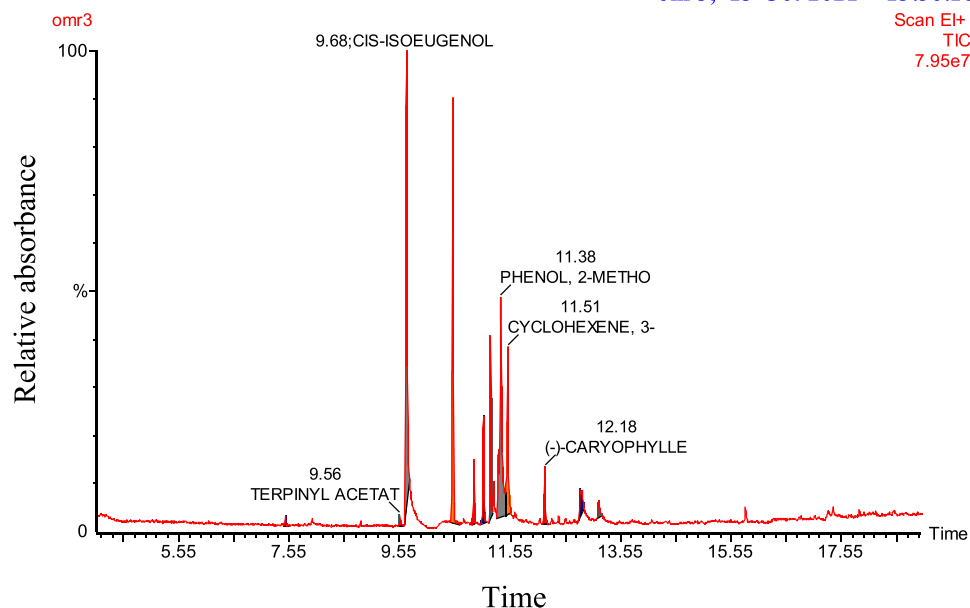


Fig. 5 Antioxidant activity of HF6H extract with different concentrations assessed by DPPH assay. Values are mean $\pm$ standard deviation of triplicate determinations

types of cancer (Table 1). Through the GC–MS analysis of the *Salvia officinalis* essential oil extract, the presence of trans-caryophyllene showed anticancer potential against RAW246.7 and HCT-116 (El Hadri et al. 2010). Similarly, isoeugenol exhibited high cytotoxicity against human cancer cells, namely, HeLa, FemX, K562, and SKOV3 (Zarlaha et al. 2014). Farnesene was also reported to possess

significant bioactivities such as antimicrobial, anticancer, and free radical-scavenging activities (ÇELİK et al. 2014). The anticancer activity of zingiberene against HT-29 CC cells was found to be attributed to the induction of autophagy (Chen et al. 2019a). α -Curcumin is an active component of *Curcuma zedoaria*, which induced apoptosis on the human ovarian cancer and SiHa cells (Shin & Lee 2013). Additionally, it was well documented that α -humulene induces apoptosis in hepatocellular carcinoma by inhibiting Akt signaling (Chen et al. 2019b).

Antioxidants are effective against different human metabolic disorders, cell membrane alteration, and DNA degradation (Miron et al. 2019). The antioxidant activity of HF6H extract could be accredited to the molecular structure of polyphenols and their potential in delocalizing of the free radical unpaired electrons (Silva et al. 2000). These findings comply with the published data concerning phenols and polyphenols that are commonly found in the plant kingdom. Such compounds act as free radical scavenging agents through various mechanisms including blocking, suppressing, or interfering with enzymes involved in reactive oxygen species (ROS) generation, chelating transition metals and quenching free radicals to render inactive species (Wong et al. 2006). There are limitations of this study including the availability of resources to carry out more experiments in vivo and in vitro.

Fig. 6 Gas chromatography–mass spectrometry chromatogram of HF6H extract**Table 1** Compounds detected in HF6H extract and their reported biological activities in literature

Compounds	RT	Area %	MF	MW	Compound type	Botanical source	Biological activity	References
Camphor	7.49	0.430	C ₁₀ H ₁₆ O	152.23	Terpene	<i>Salvia libanotica</i>	Anticancer	Ramana et al. (2014)
Terpinyl acetate	9.56	1.700	C ₁₂ H ₂₀ O ₂	196.28	Monoterpenoid	<i>Elettaria cardamomum</i>	Anticancer	Loizzo et al. (2008)
Cis-isoeugenol	9.68	18.420	C ₁₀ H ₁₂ O ₂	164.20	Phenol	<i>Ocimum basilicum</i>	Anticancer	Zarlaha et al. (2014)
Trans-caryophyllene	10.52	21.550	C ₁₅ H ₂₄	204.35	Sesquiterpene	<i>Ocimum gratissimum</i> and <i>Cannabis sativa</i>	Attenuate pain in rat Anticancer	El Hadri et al. (2010) and Paula-Freire et al. (2014)
α-Humulene	10.90	4.620	C ₁₅ H ₂₄	204.35	Terpene	<i>Eupatorium odoratum</i>	Anticancer	Chen et al. (2019a)
Curcumene	11.07	4.950	C ₁₅ H ₂₂	202.33	Sesquiterpene	<i>Curcuma zedoaria</i>	Anticancer	Shin & Lee (2013)
Zingiberene	11.20	8.550	C ₁₅ H ₂₄	204.35	Sesquiterpene	Ginger rhizomes	Anticancer	Chen et al. (2019a)
Acetyleneugenol	11.38	17.530	C ₁₂ H ₁₄ O ₃	206.23	Phenol	<i>Acacia nilotica</i>	Antibacterial Antioxidant	Abdulhamid et al. (2021) and Kumar et al. (2014)
α-Farnesene	11.51	10.010	C ₁₅ H ₂₄	204.35	Sesquiterpene	<i>Malus domestica</i>	Antimicrobial Antioxidant Anticancer	ÇELİK et al. (2014)
Caryophyllene oxide	12.18	2.990	C ₁₅ H ₂₄ O	220.35	Sesquiterpene	<i>Syzygium aromaticum</i>	Anticancer	Fidy et al. (2016)
β-Turmerone	12.82	6.220	C ₁₅ H ₂₀ O	218.33	Sesquiterpene	<i>Curcuma xanthorrhiza</i>	Anticancer	Itokawa et al. (1985)
α-Turmerone	13.16	3.020	C ₁₅ H ₂₂ O	218.33	Sesquiterpene	<i>Curcuma longa</i>	Anticancer	Jacob & Toloue (2013)

Conclusion

The HF6H extract showed the presence of alkaloids, phenolic compounds, sterols, and terpenes. This extract

exhibited great antioxidant and anticancer potential against different cancer cell lines, and moderate cytotoxicity against non-cancer cells, rendering it a promising alternative

remedy. However, further research should be carried out to examine the toxicity of the extract on animal model.

Author contribution Conceptualization, N.A.; methodology, A.A., N.A., B.A., M.W., F.A.; software, N.A., B.A., A.A.; writing—original draft, preparation, N.A. All authors have read and agreed to the published version of the manuscript.

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Data availability All data are mentioned in the body of manuscript, tables, and figure.

Declarations

Ethics approval Not applicable.

Consent to participate Not applicable.

Consent for publication Not applicable.

Conflict of interest The authors declare no competing interests.

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